

20 m CW Transmitter — xmitter

KiCad Symbol Library Reference

Library file: **xmitter_prj/kicad/xmitter.kicad_sym**
Generated: **2026-06-04**
Symbols: **10 components defined**
Project: **20 m CW transmitter, solid-state VFO + digital keyer + vacuum-tube PA**

Symbol	Description	Package	Manufacturer	MPN
MC1496	Balanced Modulator / Demodulator	DIP-14	onsemi	MC1496P
MCP4921	12-bit SPI DAC, Single Channel	DIP-8	Microchip Technology	MCP4921-E/P
TL072	Dual Low-Noise JFET OpAmp (3...)	DIP-8	Texas Instruments	TL072CP
ADS1115	16-bit 4-Channel I ² C ADC	SOP-10	Texas Instruments	ADS1115DGST
ST7735_Display	1.8" SPI TFT Display Breakout	18-pin header	Adafruit Industries	Adafruit #498
W5500	Hardwired TCP/IP Ethernet	8-pin header	WIZnet	W5500
6N137	High-Speed Optoisolator (RSMB...)	DIP-8	Lite-On	6N137
Encoder_5734	60 mm Rugged Rotary Encoder	Panel mount, screw terminal	Adafruit Industries	Adafruit #5734
Metro_ESP32S3	Adafruit Metro ESP32-S3 Module	Dual headers	Adafruit Industries	Adafruit #5500
Si5351A_Breakout	Adafruit Si5351A Clock Generator	4-pin header + STEMMA QT	Adafruit Industries	Adafruit #5640

Note: Symbols are hand-created from datasheet pinouts. Verify footprint assignments before PCB layout.

MC1496 — Balanced Modulator / Demodulator

Package: DIP-14 Manufacturer: onsemi MPN: MC1496P

#	Name	Function	#	Name	Function
1	SIG_IN+	Signal input + (AGC/modula...	8	CAR_IN-	Carrier input –
2	GADJ_A	Gain adjust A	9	NC	—
3	GADJ_B	Gain adjust B	10	CAR_IN+	Carrier input +
4	SIG_IN-	Signal input – (AGC/modula...	11	NC	—
5	BIAS	Output bias	12	OUT-	Collector output –
6	OUT+	Collector output +	13	NC	—
7	NC	—	14	VEE	Positive supply (+12 V)

Design notes & gotchas:

- VEE (pin 14) is the POSITIVE supply — counterintuitive Motorola naming convention.
- SIG_IN pins (1, 4) take the 0–10 V key-shaping voltage in this design (AGC/modulating port).
- Pins 7, 9, 11, 13 are NC — leave unconnected; no X markers required.
- MC1496P (DIP) has limited availability. SOIC MC1496DR2G is more readily stocked.

MCP4921 — 12-bit SPI DAC, Single Channel

Package: DIP-8 Manufacturer: Microchip Technology MPN: MCP4921-E/P

#	Name	Function	#	Name	Function
1	VDD	Positive supply (3.3 V)	5	~LDAC	Latch DAC output, active low
2	~CS	SPI chip select, active low	6	VREF	Reference voltage input
3	SCK	SPI clock	7	VSS	Ground
4	SDI	SPI data input	8	VOUT	DAC analog output (0 – VREF)

Design notes & gotchas:

- Connect VREF to 3.3 V rail — output range is 0–3.3 V before the TL072 gain stage ($\times 3 \rightarrow 0\text{--}9.9\text{ V}$).
- ~LDAC: pull permanently low for immediate output update on each SPI write.
- SPI command word: 0x3000 OR 12-bit data (gain=1 $\times$, unbuffered, active mode).
- Shares SPI bus with ST7735 and W5500 — each device must have its own CS line.

TL072 — Dual Low-Noise JFET Op-Amp (3 units: A, B, Power)

Package: DIP-8 Manufacturer: Texas Instruments MPN: TL072CP

#	Name	Function	#	Name	Function
1	OUT [A]	Op-amp 1 output	5	IN+ [B]	Op-amp 2 non-inverting input
2	IN– [A]	Op-amp 1 inverting input	6	IN– [B]	Op-amp 2 inverting input
3	IN+ [A]	Op-amp 1 non-inverting input	7	OUT [B]	Op-amp 2 output
4	VCC–	Negative supply (–8.2 V fr...	8	VCC+	Positive supply (+12 V fro...

Design notes & gotchas:

- Multi-unit symbol: place U?A and U?B as separate op-amp instances; U?C for power connections.
- Supply: +12 V / –8.2 V from existing rig rails — 20.2 V total, within 36 V absolute max.
- Output swings to within ~2 V of +rail: 10 V output is achievable with a +12 V supply.
- Tie unused unit: IN+ to GND via 10 k Ω , IN– to its own output. Never leave inputs floating.

ADS1115 — 16-bit 4-Channel I²C ADC

Package: MSOP-10 Manufacturer: Texas Instruments MPN: ADS1115IDGST

#	Name	Function	#	Name	Function
1	ADDR	I ² C address select	6	AIN2	Analog input 2
2	~ALERT/RDY	Alert / conversion ready, ...	7	AIN3	Analog input 3
3	GND	Ground	8	VDD	Positive supply (2.0–5.5 V)
4	AIN0	Analog input 0	9	SDA	I ² C data, bidirectional
5	AIN1	Analog input 1	10	SCL	I ² C clock

Design notes & gotchas:

- ADDR sets I²C address: GND→0x48, VDD→0x49, SDA→0x4A, SCL→0x4B.
- ~ALERT/RDY is open-drain — pull-up resistor required (4.7 kΩ to VDD).
- ADS1115IDGST is tape-and-reel (2500 min). Verify single-piece orderable PN at Digi-Key.
- MSOP-10 is 0.5 mm pitch — use a breakout board for breadboard prototyping.
- Selected over the ESP32-S3 onboard ADC, which is notoriously nonlinear.

ST7735_Display 1.8" SPI TFT Display Breakout (Adafruit #498)

Package: 10-pin header Manufacturer: Adafruit Industries MPN: Adafruit #498

#	Name	Function	#	Name	Function
1	GND	Ground	6	~TFT_CS	TFT chip select, active low
2	VCC	Supply (3.3–5 V)	7	MOSI	SPI data in
3	~RESET	Reset, active low	8	SCK	SPI clock
4	D/C	Data / Command select	9	MISO	SPI data out (SD card only)
5	~CARD_CS	SD card chip select, activ...	10	LITE	Backlight LED anode

Design notes & gotchas:

- Pin numbers match the Adafruit #498 header order — verify against board before PCB layout.
- ~CARD_CS and MISO only needed if using the onboard SD card; tie ~CARD_CS high if unused.
- LITE can be PWM-controlled for brightness, or tied directly to 3.3 V for always-on backlight.
- Shares SPI bus with MCP4921 and W5500 — separate CS lines required for each.
- Will be panel-mounted externally with a laser-cut acrylic bezel (SendCutSend).

W5500 — Hardwired TCP/IP Ethernet Controller Module

Package: 8-pin header Manufacturer: WIZnet MPN: W5500

#	Name	Function	#	Name	Function
1	~INT	Interrupt, active low, ope...	5	MISO	SPI data out
2	~CS	SPI chip select, active low	6	~RST	Reset, active low
3	SCK	SPI clock	7	3V3	Positive supply (3.3 V)
4	MOSI	SPI data in	8	GND	Ground

Design notes & gotchas:

- Currently backordered — all other subsystems can be developed and tested without it.
- Pin order varies by module vendor (Adafruit, WIZnet EVB, clones) — verify when module arrives.
- ~INT is open-drain — pull-up resistor required to 3.3 V.
- ~RST should be asserted briefly on power-up; drive from MCU GPIO or an RC reset circuit.
- Must be enclosed inside the RFI shielded box alongside the Metro ESP32-S3.

6N137 — High-Speed Optoisolator (10 MBd, open-collector)

Package: DIP-8 Manufacturer: Lite-On MPN: 6N137

#	Name	Function	#	Name	Function
1	A	LED anode	5	VO	Output, open-collector
2	K	LED cathode	6	VE	Enable, active high
3	NC	—	7	VCC	Output-side supply
4	GND	Output-side ground	8	NC	—

Design notes & gotchas:

- ⚠ ⚠ **VE (pin 6, Enable) MUST be tied to VCC — leaving it floating causes erratic or no output.**
- VO is open-collector — external pull-up resistor required (typ. 470 Ω to VCC).
- Two isolated power domains: LED side (pins 1–2) and logic side (pins 4, 7). Keep grounds separate.
- LED series resistor required on input: 270 Ω for ~10 mA at 5 V, 150 Ω at 3.3 V.
- One 6N137 per paddle input — use two devices for dit and dah isolation.

Encoder_5734 50 mm Rugged Rotary Encoder, 100 PPR (Adafruit #5734)

Package: Panel-mount, screw terminals Manufacturer: Adafruit Industries MPN: Adafruit #5734

#	Name	Function
1	A	Quadrature output A
2	B	Quadrature output B
3	C	Common (ground reference)

Design notes & gotchas:

- C (common) is the ground reference — connect to MCU ground.
- A and B need pull-up resistors (10 kΩ) or enable MCU internal pull-ups in firmware.
- 100 PPR = 400 counts/revolution with quadrature (×4) decoding in software.
- If rotation direction is reversed, swap A and B in firmware — no hardware change needed.
- ⚠ ⚠ **Poor manufacturer documentation — verify screw terminal labeling with ohmmeter before wiring.**
- Footprint is a placeholder 3-pin header; replace with screw-terminal connector for PCB.

Metro_ESP32S3

Adafruit Metro ESP32-S3 Module (Adafruit #5500)

Package: Module, dual headers Manufacturer: Adafruit Industries MPN: Adafruit #5500

#	Name	Function	#	Name	Function
RESET~	{RESET}	Reset, active low	D13	D13/SCK	GPIO / SPI SCK / LED
3V3_L	3V3	3.3 V output (left header)	VIN	VIN	5 V power input
AREF	AREF	Analog reference	GND_R	GND	Ground (right header)
GND_L	GND	Ground (left header)	GND_R	GND	Ground (right header)
D0	D0/RX	GPIO / UART0 RX	5V	5V	5 V output (USB-powered)
D1	D1/TX	GPIO / UART0 TX	3V3_R	3V3	3.3 V output (right header)
D2	D2	GPIO	A0	A0	Analog / GPIO
D3	D3	GPIO	A1	A1	Analog / GPIO
D4	D4	GPIO	A2	A2	Analog / GPIO
D5	D5	GPIO	A3	A3	Analog / GPIO
D6	D6	GPIO	A4	A4/SDA	Analog / GPIO / I ² C SDA / ...
D7	D7	GPIO	A5	A5/SCL	Analog / GPIO / I ² C SCL / ...
D8	D8	GPIO	SCK_H	SCK	SPI breakout header SCK
D9	D9	GPIO	MOSI_	MOSI	SPI breakout header MOSI
D10	D10/SS	GPIO / SPI SS	MISO_	MISO	SPI breakout header MISO
D11	D11/MOSI	GPIO / SPI MOSI	TX1	TX1	UART1 TX
D12	D12/MISO	GPIO / SPI MISO	RX1	RX1	UART1 RX

Design notes & gotchas:

- ⚠️ Verify pin order against Adafruit pinout diagram before wiring — ESP32-S3 GPIO mapping is board-specific.
- D11/MOSI, D12/MISO, D13/SCK and the right-side SPI header pins are the same physical signals.
- A4/SDA and A5/SCL are also the STEMMA QT connector signals — ADS1115 connects here via STEMMA QT cable.
- WiFi/BT must remain disabled in firmware for RFI control; entire board must be enclosed in shielded box.
- 5V output (right header) is only present when powered via USB-C; use VIN for external 5 V supply input.

Si5351A_Breakout

Adafruit Si5351A Clock Generator Breakout (Adafruit #5640)

Package: 7-pin header + STEMMA QT Manufacturer: Adafruit Industries MPN: Adafruit #5640

#	Name	Function	#	Name	Function
VIN	VIN	Power input (3.3–5 V, onbo...	CLK0	CLK0	Clock output 0 (0–3.3 V sq...
GND	GND	Ground	CLK1	CLK1	Clock output 1 (0–3.3 V sq...
SCL	SCL	I ² C clock, level-shifted, ...	CLK2	CLK2	Clock output 2 (0–3.3 V sq...
SDA	SDA	I ² C data, level-shifted, 1...			

Design notes & gotchas:

- I²C address: 0x60 (fixed on this breakout — no address select jumper on #5640).
- CLK0, CLK1, CLK2 are independent outputs, each configurable 8 kHz – 160 MHz.
- STEMMA QT connector provides I²C (VCC 3.3 V, GND, SCL, SDA) — use a JST SH 4-pin cable.
- Onboard LDO accepts 3.3–5 V; supply 3.3 V from Metro 3V3 rail for common-voltage operation.
- All three outputs share two PLL sources (PLLA, PLLB) — CLK frequencies may be interdependent.